

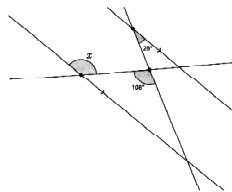
AMTI (NMTC) - 2015
GAUSS CONTEST - PRIMARY LEVEL

PART - A

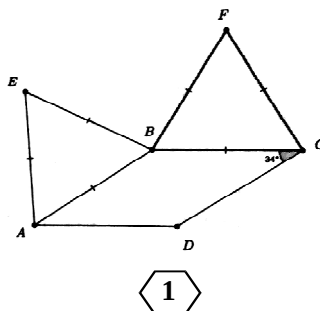
1. In the following sequence 11, 88, 16, 80, 21, 72, -, -, -, - the blanks are two digit numbers. No number in the blank ends with
(A) 1 (B) 4 (C) 6 (D) 7
2. Aruna has a piece of cloth measuring 128 cm by 72 cm. She wants to cut it into square pieces. The greatest possible size of the square that she can cut is
(A) 6 cm by 6 cm (B) 8 cm by 8 cm (C) 9 cm by 9 cm (D) 12 cm by 12 cm
3. When 26 is divided by a positive integer n , the remainder is 2. The sum of all the possible values of n is
(A) 57 (B) 60 (C) 45 (D) 74
4. Samrud, Saket, Slok, Vishwa and Arish have different amounts of money in Rupees, each an odd number which is less than 50. The largest possible sum of these amounts (in Rupees) is
(A) 229 (B) 220 (C) 250 (D) 225
5. Mahadevan used his calculator (which he rarely uses) to multiply a number by 2. But by mistake he multiplied by 20. To obtain the correct result the must
(A) divide by 20 (B) divide by 40 (C) multiply by 10 (D) multiply by 0.1
6. $a4273b$ is a six digit number in which a and b are digits. This number is divisible by 72. Then
(A) $b - 2a = 0$ (B) $a - 2b = 0$ (C) $2a - b = 4$ (D) $a + b = 13$
7. P and Q are natural numbers. If $25 \times P \times 18 = Q \times 15$. The smallest value of $P + Q$ is
(A) 61 (B) 21 (C) 41 (D) 31
8. The thousands digit in the multiplication 111111×111111 is
(A) 1 (B) 2 (C) 3 (D) 4
9. The sum of the present ages of 5 brothers is 120 years. How many years ago was the sum of their ages 80 years ?
(A) 6 (B) 7 (C) 8 (D) 9
10. Laxman starts counting backwards from 100 by 7's. He begins 100, 93, 86 Which number will not come in his countdown ?
(A) 65 (B) 30 (C) 23 (D) 15

PART - B

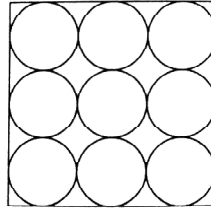
11. Jingle has six times as much money as Bingle. Dingle has twice as much money as Bingle. Pingle has six times as much many as Dingle. Pingle has _____ many times as much money as Jingle.
12. In the figure the arrowed lines are parallel. The value of x is _____.



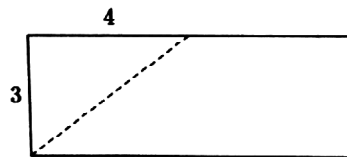
13. In the figure ABCD is a rhombus. BFC and ABE are equilateral triangles. $\angle BCD = 34^\circ$. Then $\angle EFB =$ _____.



14. In the figure, the area of each circle is 4π square units. The area of the square in the same square units is _____.



15. The maximum number of rectangles with different perimeters and an area of 216 cm^2 , if the length and breadth of each rectangle are integer multiples of 3 is _____.
16. If the previous month is July, then the month 21 months from now is _____.
17. The sum of all natural numbers less than 45 which are not divisible by 3 is _____.
18. A rectangle of dimensions 3 cm by 8 cm is cut along the dotted line shown. The cut piece is then joined with the remaining piece to form a right angled triangle. The perimeter of this triangle is _____ cm.



19. Candles A and B are lit together. Candle A lasts 11 hours and candle B lasts 7 hours. After 3 hours the two candles have equal lengths remaining. The ratio of the original length of candle A to candle B is _____.
20. A, B, C are three toys. A is 50% costlier than C and B is 25% costlier than C. Then A is _____ % costlier than B.
